

NANAVU '26 – THE 2050 KERALA HOME

OFFICIAL RULES & PARTICIPATION GUIDELINES

CONCEPT PITCHING

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1. INTRODUCTION

The NANAVU '26 Concept Pitching invites students to imagine and develop a vision for the Kerala home of 2050.

The 2050 Kerala Home

Imagine a home in Kerala that is affordable, low-carbon, climate-resilient and comfortable.

Kerala's homes of the future will need to respond to changing climatic conditions, resource constraints, evolving lifestyles, technological advancements, and the growing need for sustainable development.

The Concept Pitching challenges participants to rethink the way homes in Kerala are designed, built, operated, adapted, and experienced.

Participants are encouraged to develop innovative and practical concepts that can transform the future Kerala home into a dwelling that is:

- Affordable and accessible
- Low-carbon and resource-efficient
- Climate-resilient
- Thermally and physically comfortable
- Adaptable to changing needs
- Environmentally responsible
- Technologically enabled
- Practical and scalable (Detailed themes discussed later)

The challenge is not simply to design a futuristic house, but to envision a realistic and desirable model for housing in Kerala by 2050.

2. CONCEPT PITCHING VISION & OBJECTIVES

The Concept Pitching aims to:

- Encourage students to rethink the future of housing in Kerala.
- Explore how homes can respond to the climate and environmental challenges of 2050.
- Promote affordable and accessible sustainable housing solutions.
- Reduce the environmental and carbon footprint of residential construction.
- Encourage climate-responsive and resilient home design.
- Explore innovative, locally appropriate and low-carbon construction materials.
- Promote energy and water efficiency at the household level.
- Encourage passive design strategies that improve comfort while reducing energy demand.
- Explore circular approaches to materials, construction and household resources.
- Encourage the integration of renewable energy and appropriate technologies.
- Explore how digital technologies and AI can contribute to future housing.
- Encourage adaptable homes that can respond to changing family structures and lifestyles.
- Promote solutions that are technically feasible, economically viable and practically implementable.
- Encourage interdisciplinary collaboration among students.
- Provide a platform for students and young innovators to present future housing concepts before experts.

3. COMPETITION STRUCTURE OVERVIEW

- All teams must register for NANAVU '26 through nanavu.in.
- The presentation and evaluation of the concepts will be held at NANAVU '26.
- An abstract of the proposed concept must be submitted during registration.
- Each team must develop a concept addressing the overarching challenge of The 2050 Kerala Home.
- All submissions must be physically handed over to the organising team in the prescribed A0 poster format by 8:30 A.M. on October 2nd.

- Poster presentations will begin at 9:00 A.M. and the presentation slots will be communicated by the organising team.

4. ELIGIBILITY REQUIREMENTS

The Concept Pitching is open to students from **all branches and fields of study**.

4.1 Participant Eligibility

- Participants must be currently enrolled students of a recognized educational institution.
- Participants from different institutions may participate together, subject to the competition rules.
- Participants must provide valid institutional identification when required.
- The organisers reserve the right to verify participant eligibility at any stage.

5. TEAM COMPOSITION RULES

- Each team shall consist of **3–4 members**.
- Interdisciplinary teams are encouraged.
- A participant may be part of only one team.
- Each team must nominate one team leader.
- Teams are responsible for ensuring that all submitted information is accurate.

6. REGISTRATION RULES

- Participants must register through the official registration mechanism announced by the organisers.
- All required participant and team details must be submitted accurately.
- Each team must provide a valid team name and contact details.
- Changes to team composition after registration may be subject to approval by the organising committee.
- Incomplete or inaccurate registrations may be rejected.

- The organisers reserve the right to close registration once the maximum permitted number of teams or participants is reached.

7. THEME & CHALLENGE AREAS

THE 2050 KERALA HOME

Imagine a home in Kerala that is affordable, low-carbon, climate-resilient and comfortable.

Participants are invited to envision what a typical Kerala home could look, function and perform like in 2050.

The proposed concept may address the building itself, its materials, systems, surroundings, construction process, operation, adaptability, or the way people interact with the home.

Solutions should demonstrate how a future Kerala home can balance:

Affordability + Sustainability + Resilience + Comfort + Practicality

Participants may explore one or more of the following challenge areas.

7.1 Affordable & Accessible Housing

How can sustainable homes become affordable and accessible to a wider section of society?

Possible areas include:

- Cost-effective housing systems
- Affordable sustainable materials
- Reduction of construction costs
- Locally available materials and resources
- Modular and prefabricated construction
- Efficient construction techniques
- Incremental housing
- Adaptable housing models

- Community-based housing approaches
- Reducing lifecycle and maintenance costs

7.2 Low-Carbon Materials & Construction

How can the carbon footprint of a Kerala home be reduced?

Possible areas include:

- Low-carbon construction materials
- Recycled and reused materials
- Bio-based materials
- Locally sourced materials
- Alternative building materials
- Reduction of embodied carbon
- Low-carbon structural systems
- Material-efficient construction
- Reuse of existing building components
- Innovative construction techniques
- Appropriate traditional and vernacular building practices

7.3 Climate-Responsive & Resilient Homes

How can homes respond to Kerala's changing climate?

Possible areas include:

- Flood-resilient housing
- Heat-resilient design
- Heavy-rainfall response
- Disaster-resilient construction
- Climate-responsive site planning
- Elevated or adaptable structures
- Water management

- Passive design strategies
- Natural ventilation
- Shading and solar control
- Thermal comfort
- Climate-adaptive local construction practices

7.4 Energy-Efficient & Comfortable Homes

How can the 2050 Kerala home remain comfortable while reducing energy consumption?

Possible areas include:

- Passive cooling
- Natural ventilation
- Daylighting
- Solar shading
- Thermal insulation
- Energy-efficient building envelopes
- Renewable energy integration
- Rooftop solar systems
- Energy-efficient appliances and systems
- Smart energy management
- Reduction of operational energy demand
- Thermal comfort without excessive dependence on mechanical cooling

7.5 Water-Positive & Resource-Efficient Homes

How can future homes use resources more responsibly?

Possible areas include:

- Rainwater harvesting
- Greywater reuse
- Water-efficient fixtures

- Household water management
- Decentralised wastewater treatment
- Resource-efficient construction
- Reduction of construction waste
- Material recovery and reuse
- Household-level waste segregation
- Circular resource systems
- Lifecycle-based resource management

7.6 Circular & Regenerative Homes

Can a home become part of a circular system rather than a source of continuous resource consumption?

Possible areas include:

- Reuse of construction materials
- Recycled materials
- Design for disassembly
- Construction and demolition waste reduction
- Waste-to-resource approaches
- Material recovery
- Circular construction systems
- Composting and organic waste management
- Productive landscapes
- Integration of natural systems into housing

7.7 Adaptable & Future-Ready Homes

What should a home designed for 2050 be capable of adapting to?

Participants may explore:

- Changing family structures

- Ageing populations
- Changing work patterns
- Work-from-home spaces
- Flexible room configurations
- Modular extensions
- Universal accessibility
- Multigenerational living
- Future technologies
- Changing climatic conditions
- Easy maintenance and modification
- Design for long-term adaptability

7.8 Smart, Digital & AI-Enabled Homes

How can technology make future homes more sustainable, efficient and responsive?

Possible areas include:

- AI-assisted home design
- Smart energy management
- Digital building performance monitoring
- Automated climate control
- Water monitoring
- Smart waste management
- Digital material selection
- Building performance simulation
- Predictive maintenance
- AI-assisted sustainability assessment
- Digital twins
- Technology-enabled resource optimisation

Technology should be used where it creates meaningful value and should remain appropriate to the economic and social context of Kerala.

7.9 Kerala's Vernacular Knowledge & Future Innovation

How can Kerala's traditional knowledge contribute to the home of 2050?

Participants may explore:

- Traditional Kerala architecture
- Climate-responsive vernacular practices
- Local materials
- Traditional passive cooling techniques
- Courtyards and shaded spaces
- Traditional water-management practices
- Indigenous construction knowledge
- Adaptation of traditional techniques using modern technology
- Combining vernacular wisdom with contemporary construction

The objective is not simply to reproduce traditional architecture, but to explore how relevant principles can inform future housing.

Participants may also propose solutions outside the above areas, provided the concept clearly contributes to the vision of The 2050 Kerala Home.

A concept may address a single challenge area or combine multiple aspects into an integrated housing solution.

8. SUBMISSION REQUIREMENTS

8.1 Submission Format

Each team must submit:

- An **A0 printed poster** presenting the proposed concept.
- The required concept abstract during registration.
- Supporting materials where applicable.

9. EVALUATION CRITERIA & SCORING FRAMEWORK

Submissions may be evaluated based on the following criteria:

1. Understanding of the Challenge
2. Innovation & Creativity
3. Technical Feasibility
4. Economic Feasibility
5. Sustainability Impact
6. Climate Resilience
7. Comfort & Liveability
8. Practical Relevance
9. Scalability & Replicability
10. Implementation Potential
11. Integration of Systems
12. Presentation & Communication

The organisers may modify or adapt the evaluation parameters depending on the nature of the submitted solution.

The jury's evaluation shall be based on the information, evidence, demonstrations and presentation provided by the participating teams.

10. SUPPORTING MATERIALS

Teams may use:

- Presentations
- Physical models
- Prototypes
- Digital models
- Building simulations
- Energy simulations
- Environmental analysis
- Videos
- Demonstrations
- Drawings
- Technical illustrations
- Site and climate analysis
- Material samples
- Cost comparisons
- Lifecycle assessments

Supporting materials should directly contribute to explaining and validating the proposed concept.

11. PRESENTATION RULES

- Teams must strictly adhere to the allotted presentation time.
- All team members may participate in the presentation as permitted by the organisers.
- Visual aids may be used to communicate the proposed solution.
- Participants must be prepared for questions from the jury.
- Any claims, data, statistics or technical information should be appropriately supported.
- Participants should be able to explain the technical, environmental, social and economic basis of their concept.
- Participants must maintain professional conduct throughout the presentation.

- Offensive, discriminatory or inappropriate content is prohibited.

12. RESEARCH & INNOVATION GUIDELINES

Participants are encouraged to support their concepts with reliable evidence and research.

Teams should:

- Identify the future housing challenge clearly.
- Establish why the identified challenge is relevant to Kerala.
- Use relevant data wherever available.
- Refer to credible technical, academic, government or industry sources.
- Explain the technical basis of the proposed solution.
- Demonstrate the expected environmental, social and economic impact.
- Consider affordability and lifecycle costs.
- Consider climate and site conditions.
- Consider practical construction and implementation challenges.
- Consider lifecycle impacts wherever applicable.
- Clearly distinguish existing solutions from the team's proposed innovation.
- Avoid unsupported claims regarding cost, carbon reduction, energy savings or environmental performance.

13. SAFETY, ETHICS & COMPLIANCE

Participants must follow:

- Academic integrity.
- Laboratory and workshop safety requirements.
- Applicable institutional and professional safety standards.
- Ethical research and innovation practices.
- Accurate representation of data and results.
- Appropriate attribution of external work.
- Safe practices during prototype development and demonstration.

Participants must not misrepresent:

- Experimental results
- Technical performance
- Construction costs
- Carbon reductions
- Energy savings
- Environmental benefits
- Structural performance
- Other project information

The organisers may stop any activity considered unsafe or non-compliant.

14. INTELLECTUAL PROPERTY (IP) POLICY

- Participants retain ownership of their original ideas and intellectual property, subject to any specific agreements communicated by the organisers.
- Participants are responsible for ensuring that their submissions do not knowingly infringe the intellectual property rights of others.
- Any third-party content, technology, images, data, software or designs must be appropriately acknowledged.
- The organisers may display submitted projects for educational, promotional, documentation or event-related purposes, subject to applicable permissions.

15. PLAGIARISM & AI DISCLOSURE

- Plagiarism, copying or unauthorized use of another team's work is prohibited.
- Submitted work must represent the team's own contribution.
- Any borrowed concepts, designs, data, code, images or other materials must be appropriately acknowledged.
- Participants must disclose significant use of AI tools in the manner specified by the organisers.

- AI-generated content must not be presented as independently produced human work where disclosure is required.
- Fabrication or manipulation of data is prohibited.
- Violation of plagiarism or AI disclosure requirements may result in disqualification.

16. JURY PROTOCOLS

- Jury members shall evaluate participating teams based on the announced criteria.
- Jury members may ask questions regarding the technical, economic, environmental, social and implementation aspects of the proposed home.
- Jury discussions and deliberations shall remain confidential.
- Participants shall not attempt to influence or communicate with jury members regarding evaluation outside the designated competition process.
- Jury decisions shall be treated as final, subject to the procedures established by the organising committee.

17. DISQUALIFICATION RULES

A team may be disqualified for:

- Submission after the prescribed deadline.
- Plagiarism or unauthorized copying.
- Falsification or fabrication of data.
- Misrepresentation of project information.
- Violation of safety requirements.
- Misconduct or disruptive behaviour.
- Use of prohibited materials or methods.
- Violation of intellectual property rights.
- Failure to comply with the official rules.
- Providing false participant or registration information.
- Any action that compromises the fairness or integrity of the Concept Pitching.

The organisers reserve the right to disqualify a team at any stage if it is found to have violated the rules.

18. PRIZES

The Concept Pitching may provide:

- **Winner – ₹7,000 + ₹3,000 workshop vouchers**
- **Runner-up – ₹4,000 + ₹3,000 workshop vouchers**
- **Second Runner-up – ₹2,000 + ₹3,000 workshop vouchers**

19. CONDUCT & ANTI-HARASSMENT POLICY

NANAVU is committed to maintaining a safe, inclusive and respectful environment.

- Harassment and discrimination are not permitted.
- Participants must treat fellow participants, organisers, mentors, jury members, volunteers and guests with respect.
- Threatening, abusive or disruptive behaviour is prohibited.
- Any violation may result in removal from the event and/or disqualification.
- Participants are expected to follow the applicable institutional and venue regulations.

20. CONTACT & SUPPORT

For registration, submission, technical or event-related queries, participants may contact the official NANAVU organising team.

Official Email: nanavu.tkm@gmail.com

Website: nanavu.in

Student Coordinator:

Almina – +91 80890 99799

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IMPORTANT NOTE

The organising committee reserves the right to modify, clarify or update these guidelines where necessary. Any significant changes will be communicated to registered participants through the official communication channels.

Participation in the **NANAVU '26 – The 2050 Kerala Home Concept Pitching** constitutes acceptance of the applicable rules and guidelines.

Imagine a home in Kerala that is affordable, low-carbon, climate-resilient and comfortable.

What will it look like?

How will it be built?

How will it respond to Kerala's climate?

How will it use energy, water and materials?

And most importantly- how can it become a home that people can actually afford and live in?

Reimagine the Kerala home for 2050.